COE 379L INTRO TO MACHINE LEARNING, SECTION 14625

Do looks win votes? Correlating facial features with election results

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1. Introduction

Elections are critical to democratic systems, offering citizens the right to choose the leaders who they believe will best represent their interests, in theory, giving all citizens an equal voice in the governance of a state. However, research has suggested that non-policy-related traits, such as physical appearance, may also influence voter preferences, raising important questions about the role of unconscious biases in voter decisions. This project examines the extent to which a politician's facial structure correlates with their likelihood of electoral success.

By leveraging supervised machine learning techniques, this project seeks to quantify and analyze the relationship between appearance and election outcomes. Images of political candidates and corresponding election results will form the dataset, enabling the training of a predictive model capable of identifying patterns and correlations in facial features linked to voter preferences.

This investigation is significant for multiple reasons. First, it sheds light on a potential driver of electoral outcomes that operates independently of a candidate's policy platform or qualifications, unsettling as it may be, it challenges the idea that a politician's success relies on their merit and expertise. Second, understanding these dynamics could have broader implications for addressing biases in electoral systems and mitigating unfair advantages conferred by appearance. Finally, the novel interdisciplinary approach of this study, combining machine learning, psychology, and political science, creates a precedent for the growing importance of sophisticated computational and applied math technologies in political and social science.

2. Existing Literature

To our knowledge, the implementation of Machine Learning methods to predict electoral success based a robust numerical quantification of facial features is a novel approach to predicting election results. There have been two notable studies on this topic: A 2023 study by *Landwehr et al.* and a 2007 study by *Little et al.* Both attempt to quantify the relationship between facial features and electoral success. *Little et al.*'s work asked human participants to choose computer-generated images of faces based on the facial structure of John Kerry and George W. Bush (the major party candidates of the 2004 Presidential Election). This study found a positive correlation between the "maturity" of a face and their electoral success, however, due to the abstract and subjective nature of appearance, their results do not provide much insight outside of preliminary evidence for correlation. *Landwehr et al.*'s work built off of this approach, also using human participants (in an online setting). This study had more mathematical robustness, finding 3 facial features with the most significant impact on election results: eye ratio (the height of the eye divided by the width of the eye), eyebrow width, and the width of the mouth. While this study contained quantifiable results, it could not predict winners and losers based on the facial features it described.

These studies, while intelligent in their approach, have been subject to small datasets, human

error, unquantifiable results, and/or an inability to make future predictions. A novel Machine Learning approach makes up for these shortcomings, taking advantage of a robust numerical analysis of facial features, coupled with a very large dataset (compared to previous work), all of which contribute to the construction of a model which can make predictions of future elections.

3. METHODS

To explore the relationship between a politician's physical appearance and their electoral success, this study employed a supervised machine learning approach. The methodology can be divided into the following key stages: data collection and preprocessing, feature extraction, model development and training, and evaluation.

3.1. DATA COLLECTION AND PRELIMINARY ANALYSIS

Our dataset was constructed using portraits of candidates from the 1960 Presidential Election (John F. Kennedy vs. Richard Nixon) until the most recent 2024 Election (Donald Trump vs. Kamala Harris). There were 46 images in total, 11 of which were labeled "Winners" and the other 35 "Losers." It's important to note, that due to the nature of elections with a singular winner, there will always be more losers than winners through time. To ensure uniformity in face size and subsequent measurements, all images were standardized to a uniform resolution.

Using the Dlib (Regression Tree based) and OpenCV libraries, 68 common landmarks were mapped onto every face in the image dataset (See **Figure A.1** and **Figure A.2**). Then, the distance between every possible combination of points was calculated. Each combination was treated as a separate feature, with 2278 separate combinations per face.

In an attempt to implement the findings described in Wendher et al., the "Eye Ratio" (Height of the Eye divided by the Width of the Eye) was calculated for each candidate's respective left right eyes, and added to the feature matrix, bringing the total feature count to 2280.

To create a preliminary estimation for the most impactful features in electoral success, the absolute value of the Pearson Correlation Coefficient was calculated for each feature. The ten most correlated features had coefficients from 0.3 to 0.28. For perspective, a coefficient of 0 would signify no correlation between the feature and electoral success, while a coefficient of 1 would denote complete correlation. Eight out of the 10 most impactful features were measurements of eye width, eye height, eyebrow height, and eyebrow width (see **Figure A.3** and **Figure A.4**); which is consistent with the findings of Wendher et al. It's interesting to note that the eye ratio's themselves were not within the top 40 most impactful features, however almost every distance regarding the eyes were within the top 40. These findings show that, for our dataset, there is correlation between the shape and arrangement of the eyes/eyebrows with electoral success, much more so than other facial features.

3.2. Model Development and Training

Seeing that the correlation between facial features and election results was found to be relatively weak, it was necessary to experiment with a large variety of supervised learning models, to find the optimal one for this problem.
However, before training the models, the data had to be split into training and testing. As there are only 46 images, we decided to use 41 for training, and 5 for testing, to provide enough data for the models to train on. Furthermore, we normalized and scaled the data to prevent the picture size from affecting the predictions.
When it comes to selecting the classification models, these are the ones that were tested:

- Logistic Regression: Used for binary or multiclass classification and predicts probabilities by fitting a logistic (sigmoid) function to the data, separating classes with a linear decision boundary.
- Decision Tree: Splits data into branches based on decision rules at each node, creating a tree-like structure to classify data.
- Random Forest: An ensemble model that combines multiple decision trees, built on different subsets of the data, to improve accuracy and reduce overfitting by joining their predictions.
- Polynomial Kernel: A non-linear support vector machine that maps input data into higher-dimensional space using polynomial transformations, enabling it to classify data using curved decision boundaries.
- Radial Basis Kernel: Also a non-linear support vector machine, but it uses a Gaussian function to measure similarity, allowing it to create flexible decision boundaries based on distances between points.
- Multilayer Perceptron Neural Network: A feedforward neural network composed of multiple layers of neurons, capable of learning complex patterns by iteratively adjusting weights through backpropagation.

We performed dimensionality reduction on the data using Principal Component Analysis, to analyse the problem's linearity, shown by **Figure C.1**
We can observe that there are two separate classes, but one of them includes both winners and losers, meaning that a non-linear model is required to separate data points inside that more complex class.
This left us with five more models, and we assessed their performance with three different metrics:

- 3-fold Cross Validation
- Testing on train dataset
- Testing on test dataset

Model Performance Results

The results of our analysis across different models are presented in the table below:

Table 3.1: Performance Metrics for Various Models

Model	**Cross-Validation Accuracy**	**Train Accuracy**	**Test Accuracy**
Decision Tree	0.683150	0.975610	0.800000
Random Forest	0.710623	1.000000	1.000000
Polynomial Kernel	0.686813	1.000000	1.000000
Radial Basis Function Kernel	0.732601	0.731707	1.000000
Neural Network	0.734432	0.731707	1.000000

By these accuracies, the overall best performers are the random forest and the two kernels. However, since the kernels only output a classification, and not a probability, we decided to use the random forest, as this model does output a probability. The benefit of having this is that we can see by which proportion each person loses or wins the elections; an important observation to have.

4. Results

The key features which were previously identified from literature, and explained a large part of the success of each candidate, were tested in order to exemplify their effect on a candidate's electability. Figure B.1 exemplifies the different changes which might be observed in candidates' appearances, with their probability of victory listed underneath each image, and the color green indicating a higher probability of winning compared to losing, and red signifying the opposite. Through analysis of the scores, it can be asserted that a smaller mouth width increased a candidate's victory chances by 15%, in relative terms, from 54 to 62%. A higher eyebrows position increased electability by 6%, and smaller eyes increased electability by 73% in relative terms, when compared to an enlarged eye size.

In addition to the impact of certain features, both candidates from the 2024 presidential elections were evaluated by the model, where the model stated candidate Donald Trump's chance of victory, based on appearance, at 90%, compared to the score of 20% predicted for Kamala Harris, demonstrating a successful prediction of this year's election results.

It is of note, however, that a bias towards female candidates might be present in the model, as both female candidates which were analyzed, Hillary Clinton and Kamala Harris, had very low electability scores, 6% and 20%, respectively. This can be attributed to the larger number of male candidates in the dataset, as well as a presence of only male candidates in the winners subdivision of past data.

5. Conclusion and Outlook

In general, we obtained coherent results and were able to precisely quantify the impact of facial features on electoral success. However, as we have previously seen in *Part 4*, there remained a significant margin of error.

Firstly, we do not know for certain if a real correlation exists between facial features and election results. For instance, voters might choose the candidate they perceive as most intelligent, rational, or empathetic—qualities based on personality traits rather than facial features, which might not correlate directly with election outcomes. Even if a measurable correlation exists, several biases could lead to unreliable results.

One major source of error in our project was the dataset. Firstly, the small amount of data available to train our model posed a significant challenge. In fact, we used only 46 portraits of political leaders—11 of whom won and 35 lost. The overrepresentation of losing candidates created a greater diversity in the distances associated with the "Lost" label, increasing the likelihood of predicting a loss. Accuracy could have been improved by expanding the dataset to include images of other American candidates, not only from presidential elections but also from other institutional roles such as the Senate or Congress. This expansion would have resulted in a richer, more comprehensive, and diverse dataset.

This brings us to the second bias we identified: the lack of diversity in our dataset. Because U.S. presidential elections only have two candidates every four years, with some candidates running multiple times, much of our dataset reflected older stereotypes that have evolved over time. This bias makes the model more likely to assign higher probabilities of success to older white men, despite the fact that today's society sees increased electoral success for candidates from minorities. For instance, female candidates and individuals from diverse ethnic backgrounds now have higher chances of being elected than ever before. To address this, one solution could be to include candidates from lower-tier elections, where there is greater diversity, or to train the model using presidential candidates from other countries.

After selecting the candidates for training, we faced another challenge in extracting the data—specifically, creating landmarks and measuring distances. The third bias lies in the angle and head position of the candidate in their photograph. Different landmarks can result depending on the candidate's gaze direction, as our dataset is based on the distances between points. Moreover, facial expressions significantly influence the results. For instance, smiling increases mouth width and decreases the eye ratio, altering the pairwise distances between landmarks. To mitigate this issue, we could have conducted a more thorough search for portraits in which candidates were looking straight ahead. Alternatively, we could have increased the number of landmarks by using MediaPipe's library, which generates 468 landmarks instead of Dlib's 68, providing much higher accuracy. Finally, using ratios instead of distances could have offered additional insights into each candidate's facial structure.

The final bias we identified was that election outcomes heavily depend on the opposing

candidate. Our model assumes the existence of an "ideal" face or set of landmark distances, which is not necessarily realistic, as election outcomes are often influenced by the relative characteristics of the candidates. To address this, additional data about who each candidate won or lost against could have been included.

A. Facial Landmark Figures

This appendix includes visual representations of the facial landmarks used in the dataset.

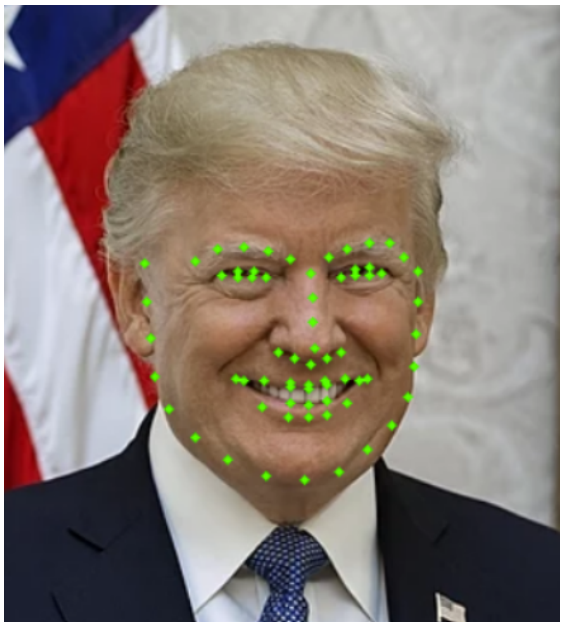


Figure A.1: *A Portrait of Donald Trump with 68 Facial Landmarks mapped using the Dlib and OpenCV libraries.*

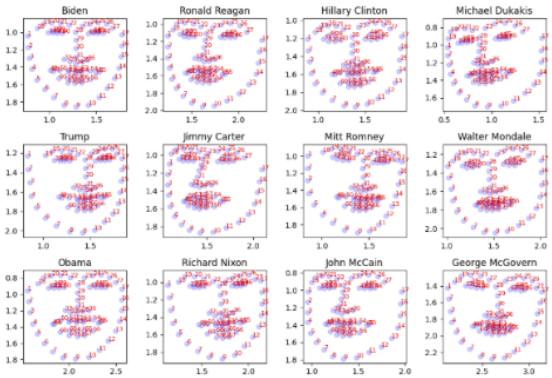


Figure A.2: *Example of Common Facial Landmarks shown for 12 candidates within the dataset.*

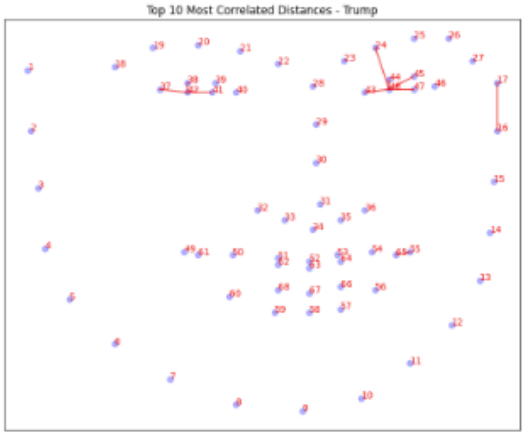


Figure A.3: *The 10 most impactful distances on electoral euccess plotted using the facial landmarks of Donald Trump*

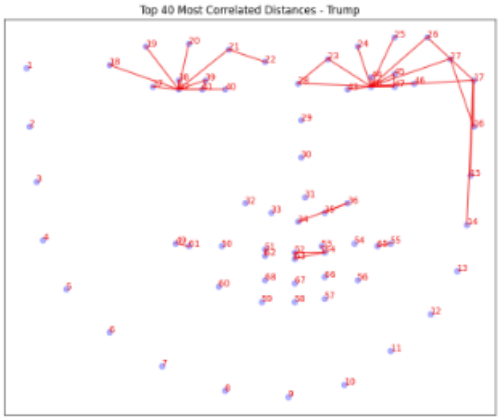
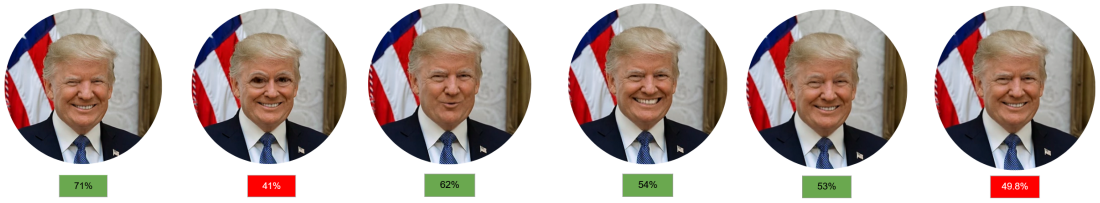


Figure A.4: *The 40 most impactful distances on electoral euccess plotted using the facial landmarks of Donald Trump*

B. Predictions

This appendix includes visual representations images from our dataset and their respective predictions.



(a) Eye Size Effect (b) Mouth Width Effect (c) Eyebrows Position Effect

Figure B.1: Effect of different changes in candidate appearance, as measured by their probability of victory

C. Analysis of the results

This appendix includes visual representations of the accuracy and performance of our model.

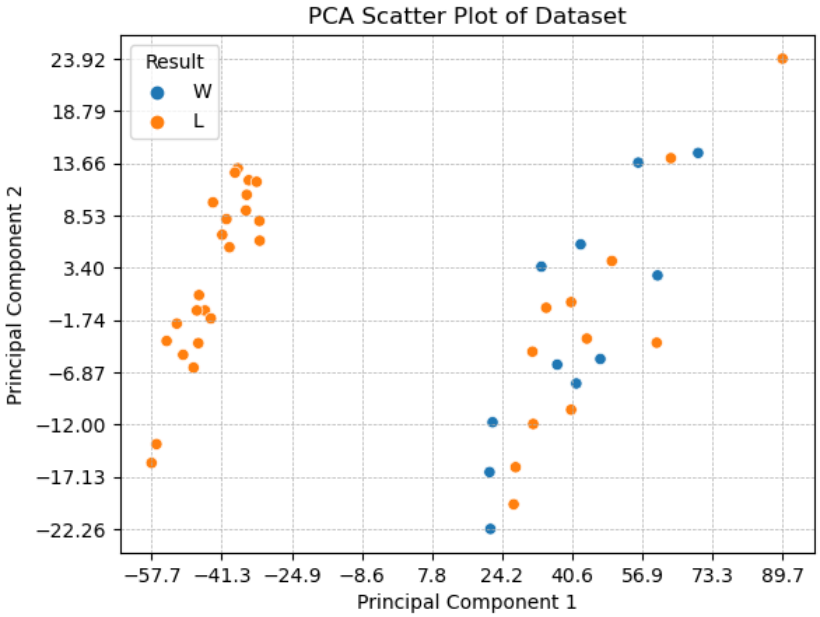


Figure C.1: PCA analysis of full dataset.